

Mini Threat Intelligence Data Warehouse using URLhaus + ThreatFox

Student

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Chosen type of project: **Data Warehouse**

Datasets / sources

URLhaus: <https://urlhaus.abuse.ch/browse/> for malicious URLs feed

ThreatFox: <https://threatfox.abuse.ch/browse/> for IOC feed

RIPEstat Data API <https://stat.ripe.net/docs/getting-started/what-is-ripestat> for the ASN/IP address enrichment

Description

I want to build a small Data Warehouse for Threat Intelligence indicators (URL, domain, IP) by integrating two public sources: URLhaus and ThreatFox. I will collect the raw data in two separate staging areas (one per source). Then I will normalize the data into a common format (indicator type/value, time bucket and tags) and apply a simple deduplication rule based on source and day. If the APIs allow it, I would like to use the RIPEstat Data API (or similar) to enrich the collected data.

After that, I will design the conceptual model using DFM and implement a star schema in PostgreSQL with fact and dimension tables. Finally I will run a few OLAP-style analyses such as: trends over time, overlap between the two sources, top TLDs/tags and (when available) online vs offline evolution.

The ETL pipeline will be kept modular so that, in the future, it can be extended with extra enrichment sources (e.g. paid services like Shodan, Censys, Spur) without changing the core model.